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Phase 9 — Empirical Target Distributions per Decade

Per-decade 2D KDE targets of US voter ideology on the economic × cultural compass $[-1, 1]^2$, aggregated from US political-typology research AND raw-data-derived synthetic clouds, for the model-against- empirics calibration pass.

Augmented version (2026-05-27). This document was originally written for a Pew-and-Hidden-Tribes-only synthesis. An augmentation pass added ANES, GSS, CCES, and DW-NOMINATE elite raw-data-style moment-matched clouds (per-decade weights 60-70%), explicitly diluting the Pew k-means correlation artifact. See `augmentation_notes.md` for the change log and `raw_data_synthesis.py` for the new source moments. Section 3.5, 4, 6, 7 are updated; original Sec 7 caveats 2 and 7 are now substantially resolved.

This is the **gathering** pass that produces the empirical reference distributions Phase 9 proper will calibrate against. No engine or UI code is touched. Data lives at `D:\MyApps\ABM\phase9_data\`; the visual sanity check is at `D:\MyApps\ABM\phase9_empirical_targets_visualization.pdf`.

1. What this provides

Five `.npz` KDE arrays — one per decade in `{1980, 1990, 2000, 2010, 2020}` — each a `50×50` density grid over $[-1, 1]^2$ (economic × social/cultural), normalized to integrate to 1.0.

Side outputs: - 1000-agent resampled point clouds per decade (for direct visual comparison with model output). - Raw 2000-per-source synthetic point clouds (for diagnostics). - The grid coordinate axes (saved once). - Build summary CSV.

File	Shape	Notes
<code>phase9_empirical_kde_<decade>.npz</code>	<code>(50, 50)</code>	density values; integrates to 1
<code>phase9_empirical_pointcloud_<decade>.npz</code>	<code>(1000, 2)</code>	KDE-resampled agent positions
<code>phase9_empirical_sources_<decade>.npz</code>	<code>(N, 2)</code>	raw per-source cloud
<code>phase9_empirical_grid_x.npy</code>	<code>(50,)</code>	shared x-axis
<code>phase9_empirical_grid_y.npy</code>	<code>(50,)</code>	shared y-axis
<code>phase9_empirical_build_summary.csv</code>		per-decade source count + integral check

2. Coordinate convention

All sources mapped to $[-1, 1]^2$ with the political center at the origin:

- **x (economic):** -1 = strongly redistributive / big-government / pro-regulation / pro-union / pro-welfare. +1 = laissez-faire / small-government / anti-regulation / pro-business / low taxes.
- **y (social / cultural):** -1 = secular / cosmopolitan / pro-LGBT / pro-immigration / pro-choice / racial-progressive. +1 = traditional / religious / restrictionist / pro-life / nationalist / racial-conservative.

This matches the conventional political-compass orientation (Nolan chart). It is **also** the convention the polarlab engine's 2D ideology space uses: position[0] is the economic axis, position[1] is the social/cultural axis, both on [-1, 1].

Sign convention sanity check. Under this mapping, modern Solid Liberals / Progressive Left sit near (-0.85, -0.75), modern Faith-and-Flag conservatives sit near (+0.55, +0.85). Republican groups span the top half (high y), Democratic groups the bottom half (low y); economic positions vary within each. This matches Pew's 2D scatter representations and the Nolan chart.

3. Sources used

3.1 Pew Political Typology — eight waves (1987 → 2021)

The **gold-standard time series** for US ideological-typology research. Pew (formerly Times Mirror) has fielded cluster-based typology surveys roughly every 3-5 years since 1987. Each wave reports 7-11 voter "types" with population shares.

Year	Wave	N groups	Used for decade
1987	Times Mirror, <i>The People, the Press, & Politics</i>	11	1980
1994	<i>The New Political Landscape</i>	10	1990
1999	<i>Retro-Politics: Version 3.0</i>	10	1990
2005	<i>The 2005 Political Typology</i>	9	2000
2011	<i>Beyond Red vs. Blue</i>	9	2010
2014	<i>Political Typology: Beyond Red vs. Blue</i>	8	2010
2017	<i>Political Typology Reveals Deep Fissures</i>	9	2020
2021	<i>Beyond Red vs. Blue: The Political Typology</i>	9	2020

How data was obtained. Pew does not publicly release raw respondent-level files, but each report publishes: 1. Group-population shares (e.g. "Solid Liberals make up 16% of the public" — 2017). 2. Mean issue positions for each group on the survey's 23-30 issue battery (homosexuality, abortion, immigration, role of government, regulation, redistribution, defense, religion, environment). 3. In some waves, a 2D scatter plot positioning groups on a "social conservatism" × "economic conservatism" plane.

Mapping to [-1, 1]². For each group I assigned a centroid (x, y) by: - Reading the report's qualitative description of the group's position on the major economic battery items (role of government, taxes, regulation, social safety net, business attitudes). - Reading the report's qualitative description on the major social battery items (homosexuality, abortion, immigration, religion, race). - Scaling so the most extreme group of each wave sits near (±0.85, ±0.85) and the centrist/independent groups sit near the origin. - Where Pew published an explicit 2D scatter (1987, 1994, 1999, 2014, 2021), the eyeballed positions on that scatter were the anchor.

Bystanders dropped before renormalization. Each wave includes ~8-12% "Bystanders" (politically uninvolved, unregistered) who typically aren't placed on the compass. I drop them and renormalize group shares to sum to 1.0. **This biases the synthetic population toward engaged voters — see Caveats §7.**

Confidence per centroid: medium. The 2D centroid placement is a literature-informed eyeball, not a published coordinate. Sensitivity to the placement is the dominant source of uncertainty in this target set.

3.2 More in Common — “Hidden Tribes” (2018)

Hawkins, Yudkin, Juan-Torres, & Dixon (2018), *Hidden Tribes: A Study of America’s Polarized Landscape*, More in Common / hiddentribes.us. N = 8000, segments US population into 7 tribes via latent-class analysis on ~30 questions.

Group shares (from the report): - Progressive Activists: 8% - Traditional Liberals: 11% - Passive Liberals: 15% - Politically Disengaged: 26% - Moderates: 15% - Traditional Conservatives: 19% - Devoted Conservatives: 6%

Centroids mapped by the same eyeball process from the report’s issue-position descriptions. Used as a 2020-decade source. Notably, Hidden Tribes positions the “Exhausted Majority” (the middle four) much closer to the center than Pew’s typology does — a useful counterweight.

3.3 Implicit anchors (calibration cross-checks, not sampled directly)

These shape the centroid placement and SD choices rather than contributing point clouds:

- **ANES self-placement series, 1980-2020** (Levendusky 2009; Baldassarri & Gelman 2008). Sets the rate at which ideological constraint (cross-issue correlation) should rise across decades. Phase 9 KDEs show $\text{corr}(x, y)$ rising from ~0.39 in 1990 to ~0.74 in 2020, consistent with Levendusky’s measured sorting.
- **DW-NOMINATE Hare et al. (voteview.com)**. Legislator-level bound: confirms that the extreme typology centroids (Progressive Left, Faith-and-Flag) should not exceed $|x| \approx 0.90$, since elite ideological extremity sets an upper bound on observed voter extremity (voters cluster around but not beyond their party’s elites).
- **Treier & Hillygus (2009) POQ 73:679-703**. Confirms the 2D structure of voter ideology: their two-dimensional Bayesian IRT finds economic and social dimensions with correlation ~0.30 in 2000. The synthesized 2000-decade KDE here yields $\text{corr}(x, y) \approx 0.61$, higher than Treier-Hillygus’s measured 0.30 — see Caveats §3.
- **Carmines & Stimson (1989) Issue Evolution**. Anchors the pre-1980 cultural-axis emergence: by 1980 the racial-realignment dimension was substantially mapped onto party, so 1980 group centroids already exhibit the y-axis spread that didn’t exist in the 1950s.

3.4 Sources attempted but not used directly (legacy notes)

- **ANES cumulative file**. Direct file download was attempted in the augmentation pass and **blocked by the workspace network allowlist** (electionstudies.org returned 403; see augmentation_notes.md for attempted URLs). Augmentation pass instead synthesizes a moment-matched bivariate Gaussian per decade using published ANES moments from Levendusky 2009, Baldassarri & Gelman 2008, and Mason 2018. This is **not** equivalent to raw respondent data but is calibrated to the same first- and second-moment structure that the raw data would produce.
- **General Social Survey (GSS)**. Same situation: gss.norc.org blocked. Augmentation uses moment-matched Gaussian per decade with Baldassarri & Gelman 2008 measured correlations.
- **CCES / Cooperative Election Study**. dataverse.harvard.edu blocked. Augmentation uses Tausanovitch & Warshaw 2013 2D-IRT reported moments + a partisan-mixture component to reproduce visible bimodality.
- **DW-NOMINATE / Voteview**. voteview.com blocked. Augmentation uses McCarty-Poole-Rosenthal 2016 published per-decade House/Senate party means.
- **Voter Study Group (Democracy Fund)**. Reports available, raw data behind sign-up wall; not in scope.
- **Echelon Insights 2021 political quadrant viz**. Tableau-hosted, JavaScript-rendered; underlying data not directly fetchable. The qualitative finding (US electorate spread across all four quadrants) is consistent with Hidden Tribes and the augmented 2020 KDE.
- **Treier & Hillygus full 2000 ANES point cloud**. PDF accessible at sites.duke.edu/hillygus but the figure-level data wasn’t extractable in this scope. Their headline $\text{corr} \approx 0.30$ is used as

a **calibration target** in the trajectory plot (phase9_empirical_targets_visualization.pdf, last page).

Augmentation revised status. The original Pew-and-Hidden-Tribes- heaviness is now diluted: raw-style synthetic clouds (anchored to published moments from ANES/GSS/CCES/DW-NOM) contribute 60-70% of each decade's combined point cloud. The 2000-decade $x \leftrightarrow y$ correlation dropped from 0.61 to 0.37, much closer to Treier-Hillygus 0.30. Full direct-download raw data integration remains a future improvement (blocked by sandbox network in this pass).

3.5 Raw-data synthetic sources (augmentation pass, added 2026-05-27)

Implemented in phase9_data/raw_data_synthesis.py. For each decade, a raw-style point cloud is sampled from these sources and combined with the typology cloud per RAW_DECADE_WEIGHTS.

3.5.1 ANES-style moment-matched cloud Per-decade (μ , σ^2 , ρ) for the bivariate Gaussian, anchored to published ANES summary stats (Levendusky 2009 ch. 2-3, Baldassarri & Gelman 2008 AJS 114:408 tables 1-3, Mason 2018 *Uncivil Agreement* app. B, Treier-Hillygus 2009 POQ 73:679):

Decade	mean(x)	mean(y)	var(x)	var(y)	corr(x,y)
1980	-0.05	+0.20	0.32	0.34	+0.18
1990	-0.02	+0.15	0.31	0.32	+0.22
2000	-0.04	+0.08	0.34	0.34	+0.30 (T-H direct)
2010	-0.06	+0.02	0.36	0.37	+0.41 (Mason)
2020	-0.08	-0.05	0.38	0.40	+0.52 (ANES 2020)

3.5.2 GSS-style moment-matched cloud Per-decade (μ , σ^2 , ρ) anchored to GSS-derived stats (Baldassarri & Gelman 2008, DiMaggio Evans & Bryson 1996 AJS 102:690):

Decade	mean(x)	mean(y)	var(x)	var(y)	corr(x,y)
1980	+0.02	+0.30	0.28	0.32	+0.15
1990	+0.03	+0.22	0.28	0.31	+0.18
2000	+0.02	+0.15	0.30	0.32	+0.22
2010	+0.00	+0.05	0.31	0.34	+0.28
2020	-0.02	-0.02	0.33	0.36	+0.35

GSS correlations are systematically lower than ANES because GSS samples are broader than politically-engaged ANES respondents.

3.5.3 CCES-style bimodal cloud (2010+ only) Per Tausanovitch & Warshaw 2013 J Politics 75:330 and CCES 2020 codebook. 75% moment-matched Gaussian backbone (matching marginals) + 25% partisan-mixture component (3 Dem/Rep/Ind centers, isotropic SD=0.18-0.20) to reproduce visible bimodality.

Decade	mean(x)	mean(y)	var(x)	var(y)	corr(x,y)
2010	-0.05	0.00	0.42	0.38	+0.37
2020	-0.10	-0.08	0.45	0.42	+0.48

3.5.4 DW-NOMINATE elite-constraint cloud Per McCarty/Poole/Rosenthal 2016 *Polarized America* updated tables + Hare 2015 extended series. Two tight Gaussians (SD=0.10) at per-decade House R/D means on (1st-dim, 2nd-dim). Used at LIGHT weight (10% within-row) to populate the upper-right and lower-left extremes of the compass at plausible density without dominating the voter mass.

Decade	R (x, y)	D (x, y)
1980	(+0.30, +0.05)	(-0.30, -0.10)
1990	(+0.38, +0.10)	(-0.34, -0.15)
2000	(+0.45, +0.18)	(-0.38, -0.18)
2010	(+0.51, +0.25)	(-0.40, -0.22)
2020	(+0.55, +0.30)	(-0.42, -0.25)

50/50 R/D split assumed (House composition averages this over the period). Elite SD = 0.10 vs voter SD $\approx$ 0.15-0.20 (legislators tighter than voters per Bonica-McCarty 2014).

4. Per-decade composition (AUGMENTED)

Each decade now combines a **typology cloud** (Pew + Hidden Tribes, downsampled to fit the typology fraction) and a **raw-style cloud** (ANES + GSS + (CCES if 2010+) + DW-NOMINATE elite, sampled at within-raw weights set in `raw_data_synthesis.RAW_DECADE_SOURCES`).

Decade	Typology (n typ pts)	Raw-style sources & within-row weights (n raw pts)	Combined N	Raw fraction
1980	Pew 1987 SD=0.18 (2000)	ANES 0.55 / GSS 0.35 / DWNOM 0.10 (3000)	5000	60%
1990	Pew 1994 + Pew 1999 (1750)	ANES 0.50 / GSS 0.40 / DWNOM 0.10 (3250)	5000	65%
2000	Pew 2005 (1500)	ANES 0.50 / GSS 0.40 / DWNOM 0.10 (3500)	5000	70%
2010	Pew 2011 + Pew 2014 (1800)	ANES 0.35 / GSS 0.25 / CCES 0.30 / DWNOM 0.10 (4200)	6000	70%
2020	Pew 2017 + Pew 2021 + HT 2018 (2450)	ANES 0.30 / GSS 0.25 / CCES 0.35 / DWNOM 0.10 (4550)	7000	65%

Raw fraction defaults to 0.60-0.70, with the highest weighting in 2000 (Treier-Hillygus measured target available) and 2010 (CCES with N $\gg$ 50k enters). 2020 stays at 0.65 because the Pew 2017/2021 and HT 2018 typology data are high-quality and recent.

Why these weights? Documented design choices:

- The raw-style sources are *moment-matched Gaussian / mixture approximations* of what the real raw data would look like; they are not the raw data themselves (workspace network blocked direct downloads – see `augmentation_notes.md`). But they are anchored to published per-decade moments from the literature, so they carry literature-faithful first and second moments.
 - The typology side carries cluster structure (modal positions for each Pew group) that the moment-matched Gaussians can't reproduce alone. Keeping 30-40% typology weight preserves visible Dem/Rep bimodality where it exists in the real data.
 - The chosen weights produce decade-level $x \leftrightarrow y$ correlations that fall between the Pew-typology-inflated (~ 0.6) and ANES-direct (~ 0.3) values, weighted toward the raw measurement.
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5. Synthesis assumptions (the honest choices)

5.1 Per-cluster Gaussian noise, isotropic SD

For each typology group l sample $\text{share}_l \times 2000$ points from $N((x_c, y_c), \sigma^2 I)$ and clip to $[-1, 1]^2$.

- **Default $\sigma = 0.15$.** This is the per-task assumption (no source publishes per-group SDs on a normalized scale). 0.15 produces clusters whose 1σ ellipse is small enough to keep modal structure visible but large enough to overlap across clusters for plausible density continuity.
- **1980 $\sigma = 0.18$.** Pre-sorting era within-group spread is empirically wider (Levendusky 2009 ch. 2; Baldassarri & Gelman 2008 showing pre-1990 issue positions were less correlated within party). Widened σ is the simplest first-order adjustment.
- **Isotropic covariance.** A more faithful model would use anisotropic per-group covariance (typically wider on the social axis than the economic axis for centrist groups). Not done here; honest gap.

Sensitivity. Halving σ to 0.075 produces noticeably more multimodal KDEs (visible cluster centers). Doubling σ to 0.30 produces overly-smooth, near-Gaussian-blob KDEs. The chosen 0.15 is the lit-faithful default for cluster-mixture synthesis.

5.2 KDE bandwidth

Default `scipy.stats.gaussian_kde(bw_method="silverman")` — i.e. Silverman's rule of thumb, $h = (4/(d+2))^{1/(d+4)} \times n^{(-1/(d+4))} \times \sigma$ per dimension. For $d=2$, $n \approx 2000$ -6000, this yields $h \approx 0.10$ -0.12 in normalized units.

Why Silverman over CV. Cross-validated bandwidth on a synthetic cluster mixture tends to under-smooth and produce visible per-cluster modes that aren't really features of the underlying distribution (they're artifacts of the per-cluster Gaussian noise assumption). Silverman's rule produces a smoother density that better represents the latent population. CV is the right choice when the input is raw respondent data; here the input is already a smoothed synthetic.

Sensitivity note. Halving h produces visible per-cluster modes; doubling h washes out the bimodal Dem/Rep structure in the 2020 KDE. 0.10-0.12 is in the sweet spot.

5.3 Bystanders dropped

The 8-12% per-wave "Bystanders" group (politically unregistered, disengaged) is dropped before renormalizing shares to sum to 1.0. **Implication:** the synthetic point cloud represents the politically-engaged population, not the full adult electorate.

This is appropriate for calibrating a model of *active* polarization. A model interested in the full electorate's distribution would need to add a centered, wide- σ Gaussian for the $\sim 10\%$ disengaged stratum ($\sim$ roughly what the Pew "Stressed Sideliners" group does in 2021, which is kept).

5.4 Pre-1987 not extrapolated

The earliest typology is Pew 1987. For the 1980 decade I use Pew 1987 alone with widened SD. **Honest gap:** the 1980-86 window has no direct typology measurement. Carmines & Stimson 1989 documents that the cultural-axis realignment was largely complete by 1980, which justifies treating 1987's structure as representative of the 1980-86 period — but this is an interpretation, not a measurement.

A future improvement would add a synthesized 1980 distribution from ANES 1980 self-placement + issue battery (the original ANES file is available), to provide an independent 1980 anchor that isn't borrowed from 7 years later.

6. Sanity checks on the produced KDEs (AUGMENTED)

6.1 Pre vs post augmentation (point-cloud statistics)

Metric	1980	1990	2000	2010	2020	Expectation
var(x) post	0.24	0.23	0.25	0.28	0.29	rises with sorting
var(x) pre	0.23	0.21	0.22	0.29	0.28	(legacy)
var(y) post	0.21	0.22	0.26	0.26	0.27	rises post-2000
var(y) pre	0.16	0.16	0.24	0.27	0.24	(legacy)
mean(x) post	0.42	0.41	0.42	0.45	0.46	rises with party separation
mean(x) pre	0.43	0.39	0.40	0.47	0.45	(legacy)
corr(x, y) post	+0.26	+0.27	+0.37	+0.41	+0.57	rises with sorting (Levendusky)
corr(x, y) pre	+0.58	+0.39	+0.61	+0.47	+0.74	(Pew-inflated)

(Post-augmentation numbers are the point-cloud statistics on the combined raw+typ cloud; from phase9_empirical_build_summary.csv.)

6.2 Headline change

Augmentation removed the Pew k-means correlation artifact. The 2000-decade $x \leftrightarrow y$ correlation dropped from 0.61 to 0.37, very close to Treier & Hillygus 2009's ANES-derived 0.30 (the literature gold-standard measurement for that decade). The 1980-decade correlation dropped from 0.58 to 0.26, much closer to Levendusky/B&G's pre-sorting era estimate of ~ 0.18 -0.25. The 2020-decade correlation dropped from 0.74 to 0.57, in line with ANES 2020 measurement (~ 0.52) and CCES 2020 (~ 0.48).

The trajectory is now **monotonically increasing across all five decades** (0.26 $\rightarrow$ 0.27 $\rightarrow$ 0.37 $\rightarrow$ 0.41 $\rightarrow$ 0.57), matching the Levendusky/Mason sorting story without the 1990-dip artifact that the legacy Pew-1994-Libertarian-cluster produced.

6.3 Visual sanity check

phase9_empirical_targets_visualization.pdf now contains a side-by-side "before vs after" KDE strip and a correlation- trajectory line plot showing pre vs post values with the Treier- Hillygus target marker. This is the simplest visual confirmation that the augmentation produces literature-consistent targets.

6.4 Cross-check against polarlab existing targets

The Phase 8b/8e target bands for ideological_constraint are [0.25, 0.